Modeling and Validation of a Soccer Ball's Three-Dimensional Flight to the Goal

Problem B

Abstract

We compare gravity, drag, constant net force and a gravity–drag–Magnus model. M3 estimates constant acceleration from early positions; M4 fits a constant drag coefficient and a lift coefficient proportional to the spin parameter.

Five prescribed kick situations give six initial states and 24 forward flights, illustrating the roles of speed, elevation and spin direction. Three actual match penalties provide 39 original sampled frames. After conversion to penalty-mark coordinates, three selected positions per event enter 12 bounded inversions, whose continuations are compared with later measured samples. Per-penalty tables give measured fit positions, inverse initial states, predicted crossings and crossing deviations, with termination states marked. A separate same-state curved-shot calculation quantifies angular damping, shifting the crossing by 0.0616 m. Section 5 uses the competition positions as a reference example for individual parameter effects, crossings compatible with the same early positions, and 100,000 Monte Carlo trajectories with fixed mass and radius.

Keywords: Soccer ball, aerodynamic drag, Magnus effect, inverse prediction, trajectory comparison

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1 Introduction

1.1 Problem Background

A penalty kick places stringent spatial and temporal demands on the goalkeeper. The goalkeeper must reach an appropriate location at the right time, while deciding when and in which direction to move. Studies of penalty anticipation discuss this timing problem and show that shot speed and kicking technique affect the predictability of shot direction [1, 2]. Such work also considers cues from the kicker before contact; the present problem instead focuses on predicting the ball's later position from its measured positions after launch.

The path of a kicked soccer ball is not generally a straight line. Gravity bends the trajectory downwards, air resistance slows the ball, and rotation can produce a sideways or vertical aerodynamic force. Bray and Kerwin combined aerodynamic forces with equations of motion and compared the resulting trajectories with experimental free kicks [5]. Myers and Mitchell analyzed three-dimensional spinning-ball motion and the influence of drag and spin [10]; Maponi and Piombin developed a numerical framework that also accommodates wind [8].

The aerodynamic behavior is also affected by the ball's surface. Wind-tunnel measurements by Asai et al. examined the relationship between flow conditions and soccer-ball forces [11], while Paudel and Yan's simulations compared a football with realistic seam geometry against a smooth sphere [9]. These studies motivate a spherical geometric approximation together with an examination of how aerodynamic coefficients affect the predicted flight.

Measurements of real flights provide a basis for evaluating such models. Griffiths et al. tracked the three-dimensional motion and spin of a marked football in field conditions [3]. Goff and Carré used camera-recorded trajectories to estimate aerodynamic coefficients [15]; subsequent work by Goff et al. combined measurements from 29 flights to construct speed-dependent drag and lift curves [4].

In this problem, the initial position and two early positions are given, whereas the initial velocity and spin are not directly specified. We therefore use the equations of motion to find states compatible with the early observations and calculate their goal-plane crossings. The DFL open dataset combines match events and optical tracking data for professional matches [6, 7], allowing predicted crossings to be compared with later recorded positions.

1.2 Problem Restatement

We study a ball travelling from the penalty mark to a goal 7.32 m wide and 2.44 m high, 11 m away. The motion and prediction problem can be organized into three stages:

1. **Ball–foot contact.** Contact establishes launch velocity and spin. These initial conditions determine the forward examples and the unknown quantities reconstructed in Section 4.

2. **Airborne flight.** Gravity, drag and Magnus lift develop the trajectory according to the governing equations established in Section 3.

3. **Goal arrival and shot outcome.** The goal-plane crossing connects the flight to the shot result. Variation in initial motion, aerodynamic parameters and environmental conditions can produce a distribution of crossing locations; Section 5 examines these sensitivities and joint perturbations.

We first compare prescribed kicks, then estimate initial states from three real penalties and compare predictions with later recorded positions. Figure 1 summarizes this physical and computational sequence.

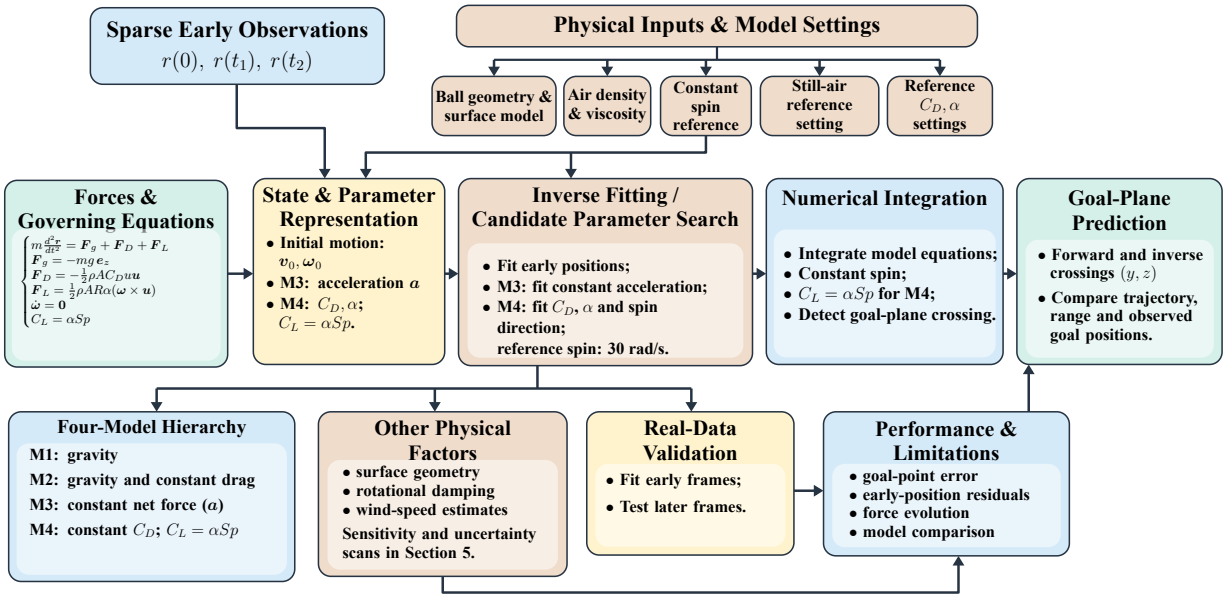


Figure 1: Three-force modeling, prescribed forward scenarios and real-penalty inverse comparison.

2 Preparation of the Models

2.1 Assumptions

- **Contact and model start.** Prescribed examples begin just after contact. Real inversions define $t = 0$ at the selected starting tracking frame; contact is represented through the initial motion.
- **Rigid spherical ball.** The ball is idealized as a rigid sphere with constant mass, radius and cross-sectional area $A = \pi R^2$; its centre of mass coincides with its geometric centre [8, 5, 10]. Empirical football coefficients represent the average aerodynamic effect of its surface. Section 6.1 discusses seams and asymmetric geometry.
- **Free-flight motion.** Gravity, aerodynamic drag and spin-induced lift act during free flight [8, 5]. The flight considered ends at the first contact with the ground, goal structure or another object; collisions, bouncing and rolling are excluded. Integration also stops at the selected goal plane.
- **Uniform gravity.** Gravitational acceleration is constant, with $g = 9.81\,\mathrm{m/s^2}$.
- **Atmospheric properties.** Air density and dynamic viscosity are spatially uniform during each flight. Aerodynamic forces are represented using empirical drag and lift coefficients.

- **Wind.** The main calculation uses still air. When wind is included, its velocity vector is spatially and temporally constant during each flight; small-scale turbulence and rapidly varying gusts are neglected. Section 6.3 examines how different wind conditions affect relative airflow and force magnitude.
- **Spin during flight.** The main model neglects aerodynamic torque, so its spin vector remains constant. Section 6.2 examines this approximation by adding rotational damping to the same initial state.
- **Observations and unknowns.** Three specified real-penalty positions enter a bounded deterministic fit, and subsequent positions supply the comparison. Released Z values are interpreted as metres and centre height. Section 4 states the search ranges for velocity, spin and force corrections; Section 5 uses the competition-position example to examine parameter effects on crossing distributions.

2.2 Notations

Table 1: Principal quantities and competition reference values.

Symbol	Meaning	Value or unit
$\boldsymbol{r}, \boldsymbol{v}$	Ball-centre position and velocity	m; m/s
$\boldsymbol{\omega}, \omega$	Spin vector and its magnitude	rad/s
m, R, A	Mass, radius and frontal area	0.43 kg; 0.11 m; πR^2
g	Gravitational acceleration	9.81 m/s^2
$\boldsymbol{u}, v$	Air-relative velocity and its magnitude	m/s
ρ	Air density	1.2 kg/m^3
$\nu, \mu = \rho\nu$	Kinematic and dynamic viscosity	1.50×10^{-5} m^2/s; Pa s
Re, Sp	Reynolds number and spin parameter	dimensionless
C_D, C_L	Drag and transverse lift coefficients	dimensionless
α_D, α_L	Multiplicative force corrections	dimensionless
$\theta, \boldsymbol{e}_L$	Spin–airflow angle and Magnus direction	rad; unit vector
X_g	Local goal-plane coordinate	11 m
$\boldsymbol{Q}, I_b$	Damping torque and moment of inertia (Section 6)	N m; kg m^2

The competition mass, radius and gravity follow the reference values of Maponi and Piombin [8]; the air density follows Asai et al. [11]. The viscosity is a declared reference air setting, with $\mu = \rho\nu$. The three match penalties use the same declared reference settings.

3 Model Establishment

The same Newtonian equations generate all four models by selecting the included aerodynamic terms. We first describe the geometry, then connect the forces to the surrounding flow and ball rotation.

3.1 Geometry and Goal Plane

The local horizontal origin is the nominal penalty mark, with x towards the goal, y lateral and z upward. The goal plane is $x = 11$ m and the opening is $-3.66 \le y \le 3.66$ m, $0 \le z \le 2.44$ m. Section 4.2 converts the measured pitch-centred coordinates.

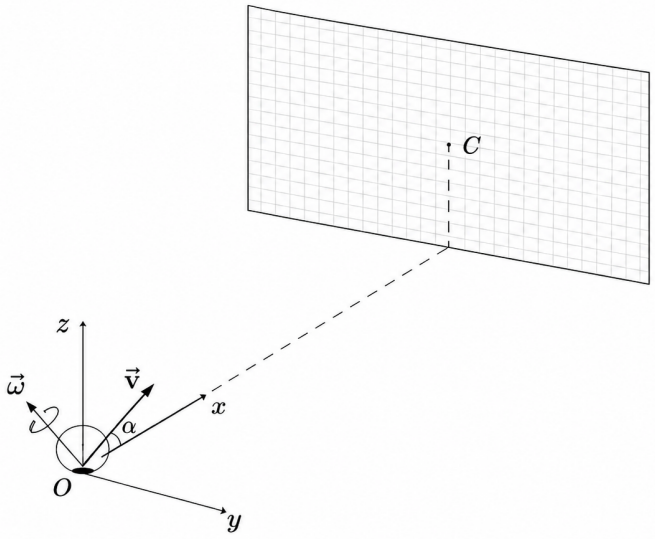


Figure 2: Penalty-kick geometry in local coordinates; O is the nominal penalty mark and C an illustrative goal-plane point.

A clean opening crossing requires $|y| \le 3.66 - R$ and $R \le z \le 2.44 - R$. Descending centre height $z = R$ defines ground contact. Every model uses these same event definitions.

3.2 Force Analysis

3.2.1 Gravity

Gravity supplies a constant downward force,

$$\boldsymbol{F}_g = -mg\boldsymbol{e}_z, \tag{1}$$

where $\boldsymbol{e}_z$ is the upward vertical unit vector. It produces downward curvature even when all aerodynamic forces are removed.

By Archimedes' principle the displaced air exerts an upward force on the ball. Its volume is $V_b = \frac{4}{3}\pi R^3$, giving

$$\boldsymbol{F}_B = \rho V_b g\, \boldsymbol{e}_z = \frac{4}{3}\pi R^3 \rho g\, \boldsymbol{e}_z. \tag{2}$$

Together, gravity and buoyancy give $\boldsymbol{F}_g + \boldsymbol{F}_B = -(m - \rho V_b)g\boldsymbol{e}_z$. For the reference parameters in the table, buoyancy is about 1.56% of the ball's weight and changes the downward acceleration from g to $g(1 - \rho V_b/m)$.

The 1.56% buoyancy scale is a small vertical correction; all four main comparisons use gravity, drag and Magnus lift as defined above.

Figure 3 illustrates the relative airflow in the ball frame and the corresponding force directions. The right-hand example takes the spin axis out of the page; the lift direction follows the cross product.

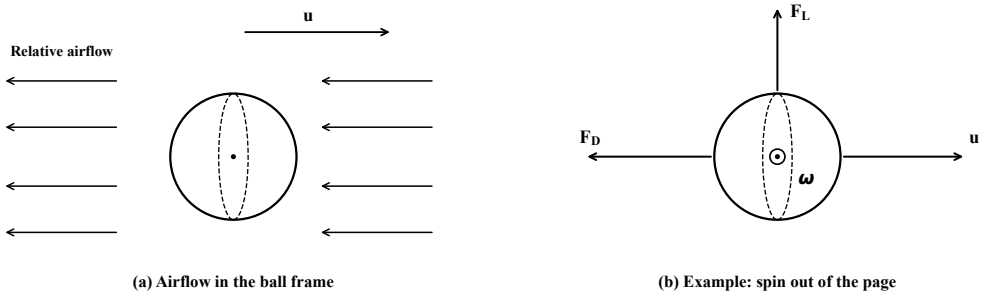


Figure 3: Relative airflow and force directions. The symbol $\odot$ denotes spin out of the page; $\boldsymbol{u}$ in the left panel is the velocity of the ball relative to the air.

3.2.2 Aerodynamic Drag

With relative velocity $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$, $u = \|\boldsymbol{u}\|$ and $A = \pi R^2$, drag is [15, 12]

$$\boldsymbol{F}_D = -\tfrac{1}{2}\rho A C_D u \boldsymbol{u}. \tag{3}$$

The drag coefficient C_D is constant during each flight.

3.2.3 Magnus Lift

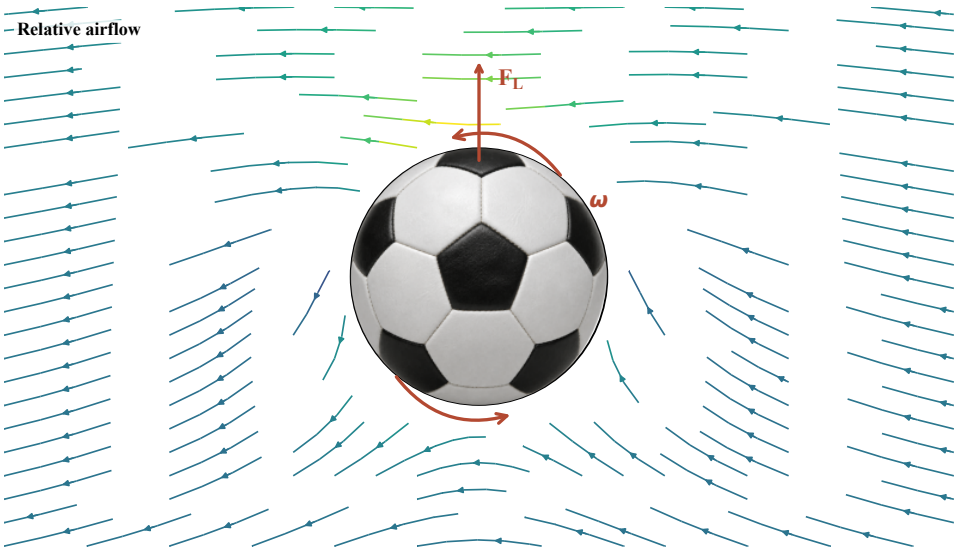


Figure 4: Idealized flow around a spinning football.

Let $Sp = R\|\boldsymbol{\omega}\|/u$ and $C_L = \alpha Sp$. Resolving the transverse component yields

$$\boldsymbol{F}_L = \tfrac{1}{2}\rho A R \alpha (\boldsymbol{\omega} \times \boldsymbol{u}). \tag{4}$$

The force vanishes at zero spin or zero relative velocity; reversing cross-product order requires reversing the sign.

3.3 Dynamical Equations

Newton's second law and angular-momentum balance give

$$m\dot{\boldsymbol{v}} = \sum \boldsymbol{F}, \qquad \frac{d\boldsymbol{L}}{dt} = \boldsymbol{Q}, \qquad \boldsymbol{L} = I_b \boldsymbol{\omega}. \tag{5}$$

With zero applied torque and constant spherical inertia, the main model is

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ \dot{\boldsymbol{\omega}} = \boldsymbol{0}. \end{cases} \tag{6}$$

Its initial conditions are

$$\begin{cases} \boldsymbol{r}(0) = \boldsymbol{r}_0, \\ \boldsymbol{v}(0) = (v_{x0}, v_{y0}, v_{z0}), \\ \boldsymbol{\omega}(0) = (\omega_{x0}, \omega_{y0}, \omega_{z0}). \end{cases} \tag{7}$$

The observed initial position fixes the origin of the flight. The six velocity and spin components determine its initial motion. In still air, drag removes mechanical energy while Magnus lift does no work along the velocity; gravity converts kinetic and potential energy during flight.

4 Model Solving

Forward prediction explains how a prescribed kick develops under different force assumptions. Inverse prediction then uses early positions from three actual penalties to estimate motion and predict the continuation. Both calculations use Eq. (6) and the same four-model hierarchy.

Table 2: Models and inverse quantities

Model	Assumption	Inverse quantities
M1	gravity	velocity (3)
M2	gravity and drag	velocity, drag coefficient (4)
M3	constant net force	velocity, acceleration (6)
M4	gravity, drag, Magnus force	velocity, spin direction, C_D, α (7)

M3 represents the combined net force. M4 uses constant C_D and α during each flight with $C_L = \alpha Sp$. Forward reference values are $C_D^* = 0.2415$ and $\alpha^* = 1.5251$ at a published reference condition [12].

4.1 Forward Prediction

4.1.1 Prescribed conditions and solution

Five kick situations are represented by six initial states. These are chosen numerical examples: power, placement, sidespin, a paired topspin/backspin comparison, and a chip. For M3, the initial M4 net force is held constant in each forward example. Each begins at $\boldsymbol{r}_0 = (0, 0, 0.11)$ m with

$$\boldsymbol{v}_0 = v_0(\cos\beta, 0, \sin\beta), \qquad \alpha_D = \alpha_L = 1. \tag{8}$$

The top- and backspin pair has the same speed and launch angle, isolating the direction of rotation. For every state, all four models are integrated by adaptive Runge–Kutta (DOP853). A forward root of $x - 11$ locates the goal crossing and a descending root of $z - R$ locates ground contact. The relative and absolute integration tolerances are 10^{-10} and 10^{-12}, with a maximum step of 0.005 s. The endpoint markers show the first goal-plane crossing or ground contact.

4.1.2 Case 1: Power shot

The ball starts at 27 m/s with an $8°$ elevation and zero spin. A high launch speed leaves little time for downward gravitational deflection. Drag removes forward momentum and increases the time required to reach the goal. With zero spin M2 and M4 have the same drag; M3 holds the initial net force constant.

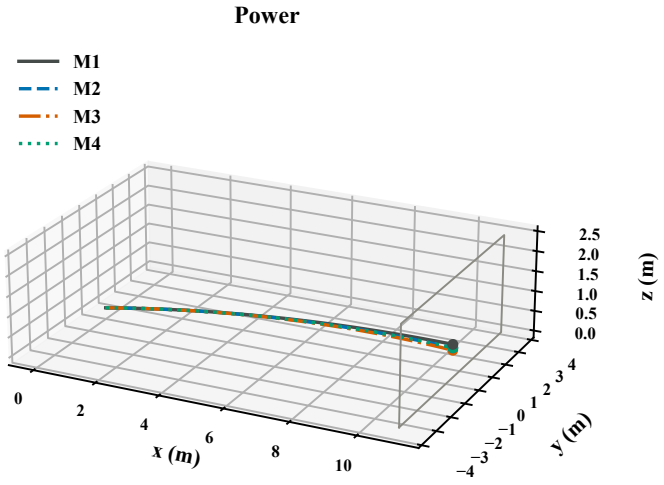


Figure 5: Power shot under the same prescribed initial condition in M1–M4. Endpoint markers denote goal crossing or ground contact.

4.1.3 Case 2: Placement shot

The ball starts at 20 m/s with an $11°$ elevation and zero spin. The lower launch speed gives gravity more time to bend the path. The comparison shows how the assumed drag coefficient changes both the travel time and crossing height. The initial direction is common to all models.

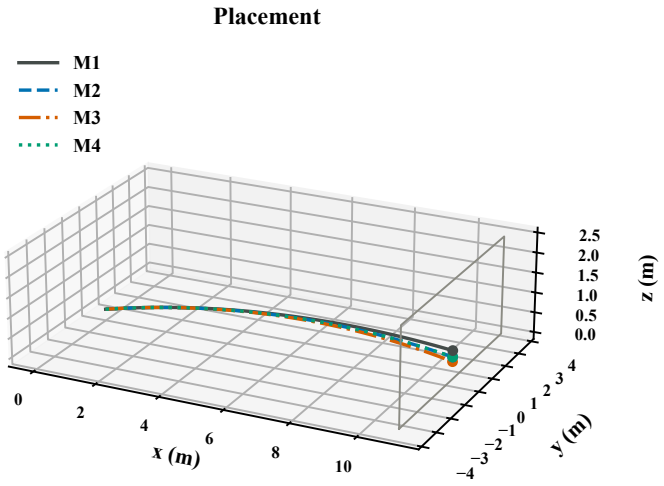


Figure 6: Placement shot with zero spin. M2 and M4 have identical drag at zero spin.

4.1.4 Case 3: Curved shot

The ball starts at 24 m/s with a 10° elevation and spin $(0, 0, 35)$ rad/s. Vertical-axis rotation creates a horizontal Magnus force. M1 and M2 preserve the initial horizontal direction; M3 holds the initial force constant while M4 evolves the Magnus force with velocity; both can bend laterally. The M4 goal crossing is $(0.8415, 0.8769)$ m at 0.5022 s. This case also supplies the reference state for the rotational-damping comparison in Section 6.2.

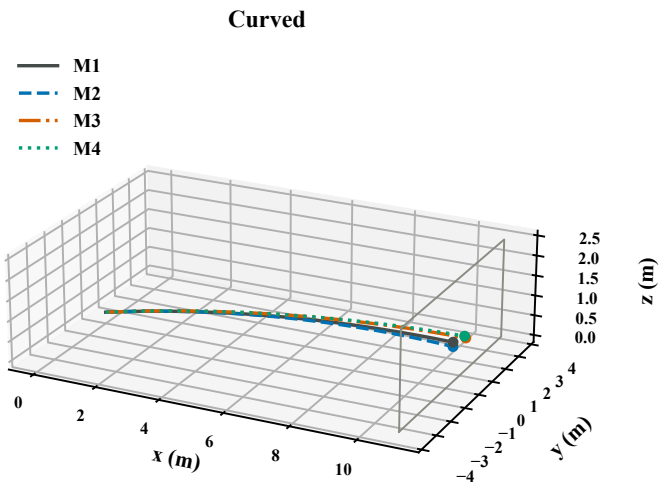


Figure 7: Sidespin redirects the ball horizontally. The four models share speed, elevation and initial spin.

4.1.5 Case 4: Topspin and backspin

Both shots start at 24 m/s and $12°$, with spin $(0, 35, 0)$ and $(0, -35, 0)$ rad/s, respectively. Changing the sign of ω_y reverses the vertical Magnus force. Topspin increases downward curvature; backspin opposes the gravitational fall and raises the trajectory. Equal launch speed and angle make this contrast visible directly. The figure shows the paths and endpoints of the models.

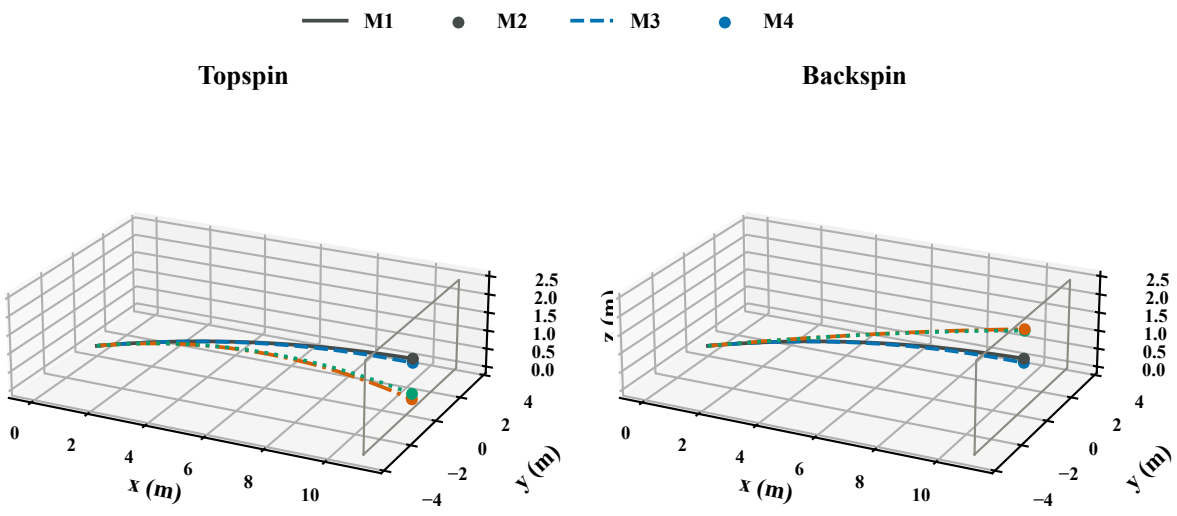


Figure 8: Matched topspin and backspin examples. Only the sign of transverse spin changes between the two panels.

4.1.6 Case 5: Chip shot

The chip starts at 14 m/s and $20°$, with spin $(0, -15, 0)$ rad/s, combining a larger elevation angle, lower speed and backspin. The initial vertical velocity builds an arc, while gravity, drag and upward Magnus lift determine how much height remains near the goal. Its slower flight lengthens the action of gravity and aerodynamic forces.

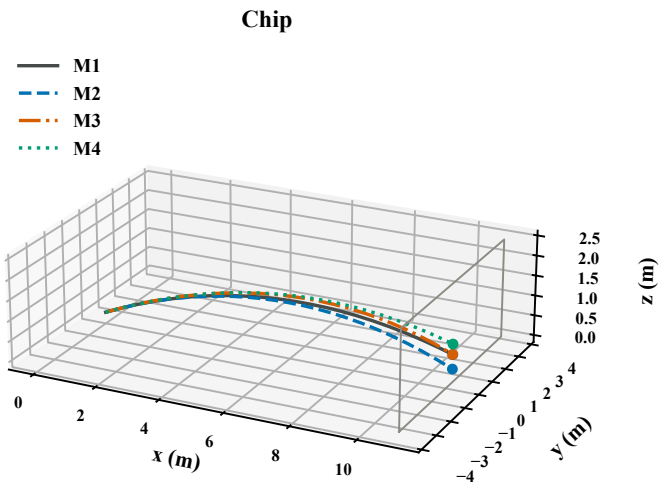


Figure 9: The prescribed chip illustrates the combined effect of launch angle, slower airflow and backspin.

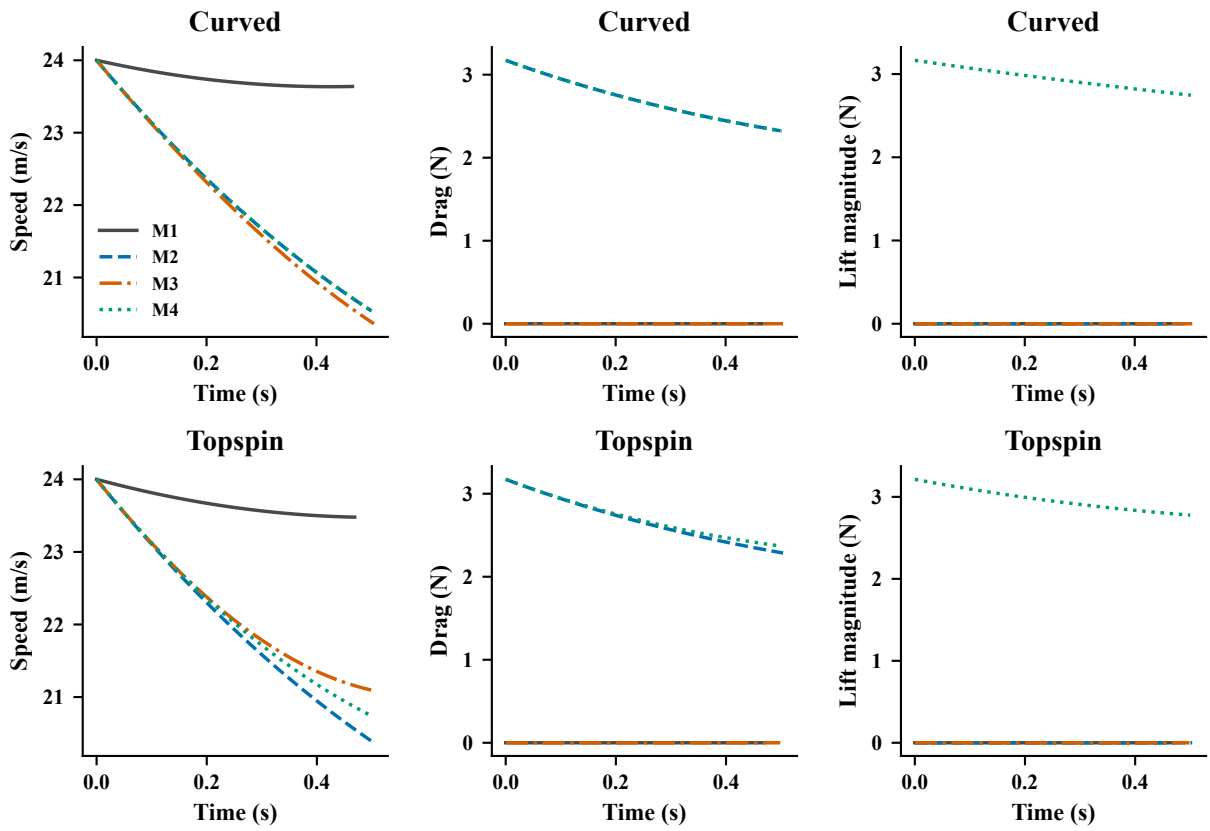


Figure 10: Speed, drag and lift magnitudes for the curved and topspin examples. Each curve ends at that model's first goal crossing or ground contact.

The force histories connect the paths to their causes. Dynamic pressure decreases as drag removes speed. A constant coefficient still gives a changing force through the v^2 factor. In M4, lift also depends on spin and relative speed. The spin direction decides whether lift acts chiefly sideways or vertically.

4.2 Inverse Prediction from Three Real Penalties

4.2.1 Measurements, coordinate system and input times

The Sportec Open DFL data associate actual match events with TRACAB Gen5 multi-camera tracking at 25 Hz [6, 7]. We use penalties by Dejan Ljubicic (J03WMX), Dawid Igor Kownacki (J03WOH), and Philipp Klement (J03WR9). The supplied CSV retains 39 source frames, respectively 15, 12 and 12. Positions are used as published; no added observation-noise model is introduced.

The source horizontal origin is the pitch centre. All three shots travel towards $X = 52.5$ m. To use the competition's penalty-mark coordinates, we apply

$$x = X - 41.5, \qquad y = Y, \qquad z = Z_{\text{original}}, \qquad x_g = 11 \text{ m}. \tag{9}$$

The nominal penalty mark is the horizontal origin, so each actual first observation retains its measured offset. Released Z values are interpreted as metres and ball-centre height. Mass, radius and air properties use the model reference settings in Section 2; the supplied source leaves per-ball mass and radius unreported.

Each inversion defines $t = 0$ at its selected source frame. J03WMX uses samples 3, 4 and 5 of the original sequence; J03WOH and J03WR9 use samples 2, 3 and 4. All three fitting times are 0, 0.04 and 0.08 s. Their time origins on the original recording are 0.08, 0.04 and 0.04 s, respectively. Table 3 gives the measured starting positions. Earlier records, repeated frames and post-goal records remain in the raw CSV. Trajectory comparison begins at the selected start, with subsequent samples used only for comparison up to the observed goal crossing. Frame-interpolated crossings are marked separately from measured frames.

Table 3: Frame counts, fitting times and measured local starting positions.

Event	Frames	Fit times (s)	$\boldsymbol{r}_0$ (m)
J03WMX	15	0.00,0.04,0.08	$(3.04, 0.55, 0.13)$
J03WOH	12	0.00,0.04,0.08	$(0.77, 0.08, 0.24)$
J03WR9	12	0.00,0.04,0.08	$(1.47, 0.49, 0.23)$

4.2.2 Bounded inversion and selection

M3 determines constant acceleration from three positions:

$$\boldsymbol{r}(t) = \boldsymbol{r}_0 + \boldsymbol{v}_0 t + \tfrac{1}{2}\boldsymbol{a}t^2. \tag{10}$$

M4 minimizes the position error over velocity, spin direction, C_D and α. Six deterministic initial guesses are used within the ranges in Table 4; the minimum-residual solution is retained.

Table 4: Component bounds and force-correction settings for inversion.

Quantity	Search interval
v_{x0} (M1, M2, M4)	$[0, 35]$ m/s
v_{y0}, v_{z0} (M1, M2, M4)	$[-35, 35]$ m/s
$\|\boldsymbol{\omega}_0\|$ (M4)	30 rad/s
Azimuth, elevation (M4)	$[-\pi, \pi], [-\pi/2, \pi/2]$ rad
$C_D/C_D^*, \alpha/\alpha^*$	$[0.5, 1.5]$

Only the product $\alpha\boldsymbol{\omega}$ enters Magnus force, so spin magnitude and α are not separately identified by positions. A reference spin magnitude of 30 rad/s is assumed. Seven fitted quantities and six later position coordinates leave multiple possible solutions.

4.2.3 Predicted trajectories, initial states and goal crossings

For each penalty we list the three measured fitting positions and observed goal crossing, followed by the initial velocity, spin information and predicted crossing from each model. M1–M3 do not estimate spin; M4 reports the reference angular-velocity vector. The deviation of a predicted crossing

from the observed crossing is

$$d_g = \sqrt{(y_{\text{pred}} - y_{\text{obs}})^2 + (z_{\text{pred}} - z_{\text{obs}})^2} \quad \text{m}. \tag{11}$$

Deviation is evaluated at each model's own goal-plane crossing. Models terminating on the ground retain their termination status, with crossing and deviation shown as "—. Open circles mark fit inputs, squares mark ground contact, and an asterisk identifies an iteration-limit candidate. All model paths stop at first ground contact.

J03WMX

Table 5: J03WMX: three measured fitting points and observed goal crossing (direct source frame).

Point	Source frame	t (s)	x (m)	y (m)	z (m)
P_0	152602	0.00	3.040	0.550	0.130
P_1	152603	0.04	4.190	0.820	0.150
P_2	152604	0.08	5.250	1.070	0.180
Goal	152610	0.3200	11.000	2.7500	0.1600

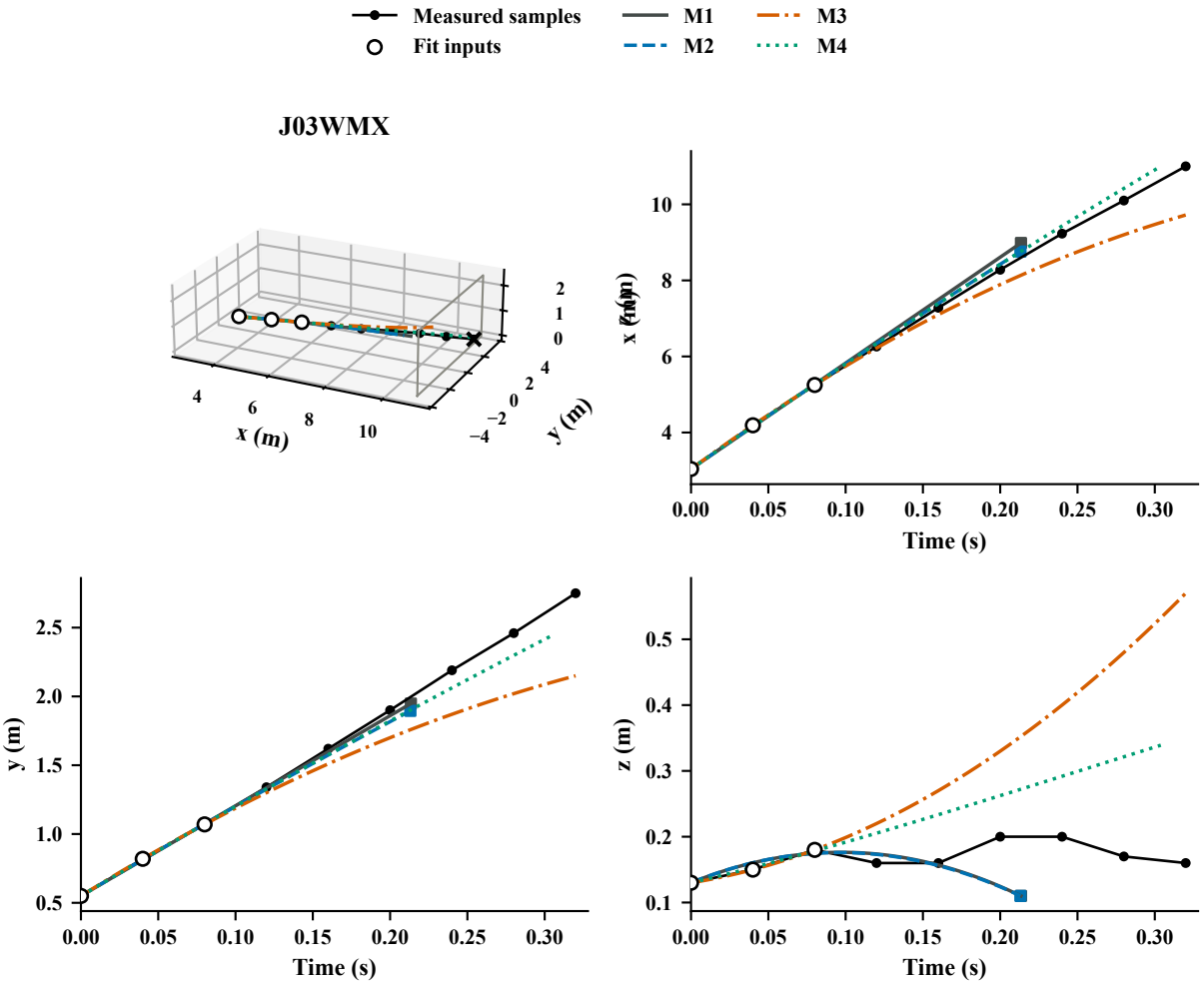


Figure 11: J03WMX: predicted and sampled trajectories from the selected start; fitting inputs are at 0, 0.04 and 0.08 s.

Table 6: J03WMX: inverse initial states and predicted goal-plane crossings. M4 uses a 30 rad/s reference spin magnitude; M1–M3 do not estimate spin.

Model	Status	y_g (m)	z_g (m)	$\boldsymbol{v}_0$ (m/s)	Spin	d_g (m)
M1	ground	—	—	$\begin{pmatrix} 27.850 \\ 6.550 \\ 0.953 \end{pmatrix}$	—	—
M2	ground	—	—	$\begin{pmatrix} 28.411 \\ 6.682 \\ 0.967 \end{pmatrix}$	—	—
M3	time limit	—	—	$\begin{pmatrix} 29.875 \\ 7.000 \\ 0.375 \end{pmatrix}$	—	—
M4	plane	2.4377	0.3390	$\begin{pmatrix} 28.423 \\ 6.667 \\ 0.551 \end{pmatrix}$	$\begin{pmatrix} 6.839 \\ -29.184 \\ 1.239 \end{pmatrix}$	0.3600

The measured low flight requires sustained vertical support from lift. M4's crossing deviation is 0.3600 m. Ground contact precedes the goal for M1, M2.

J03WOH

Table 7: J03WOH: three measured fitting points and observed goal crossing (linear interpolation).

Point	Source frame	t (s)	x (m)	y (m)	z (m)
P_0	128955	0.00	0.770	0.080	0.240
P_1	128956	0.04	1.990	0.340	0.460
P_2	128957	0.08	3.180	0.620	0.650
Goal	128964–128965	0.3643	11.000	2.9188	1.1789

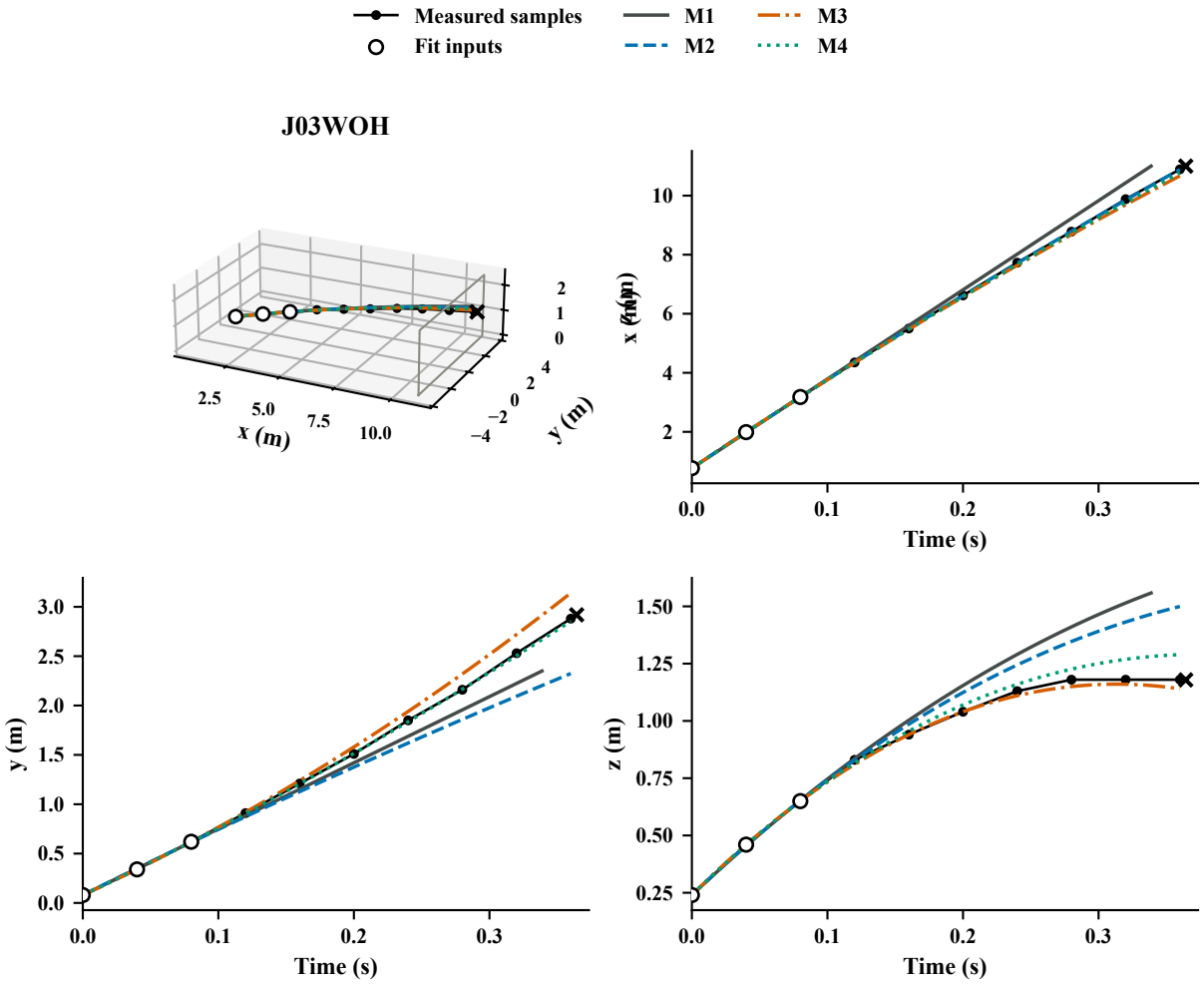


Figure 12: J03WOH: predicted and sampled trajectories from the selected start; fitting inputs are at 0, 0.04 and 0.08 s.

Table 8: J03WOH: inverse initial states and predicted goal-plane crossings. M4 uses a 30 rad/s reference spin magnitude; M1–M3 do not estimate spin.

Model	Status	y_g (m)	z_g (m)	$\boldsymbol{v}_0$ (m/s)	Spin	d_g (m)
M1	plane	2.3496	1.5583	$\begin{pmatrix}30.200\\6.700\\5.553\end{pmatrix}$	—	0.6841
M2	plane	2.3494	1.5059	$\begin{pmatrix}30.794\\6.831\\5.658\end{pmatrix}$	—	0.6567
M3	plane	3.2890	1.1262	$\begin{pmatrix}30.875\\6.250\\5.875\end{pmatrix}$	—	0.3739
M4	plane	2.9344	1.2906	$\begin{pmatrix}30.857\\6.408\\5.819\end{pmatrix}$	$\begin{pmatrix}-6.829\\9.510\\27.621\end{pmatrix}$	0.1127

The rising measured arc connects vertical launch velocity with the subsequent action of gravity and lift. M4's crossing deviation is 0.1127 m.

J03WR9

Table 9: J03WR9: three measured fitting points and observed goal crossing (linear interpolation).

Point	Source frame	t (s)	x (m)	y (m)	z (m)
P_0	176244	0.00	1.470	0.490	0.230
P_1	176245	0.04	2.570	0.850	0.340
P_2	176246	0.08	3.640	1.200	0.410
Goal	176253–176254	0.3695	11.000	3.0697	0.4657

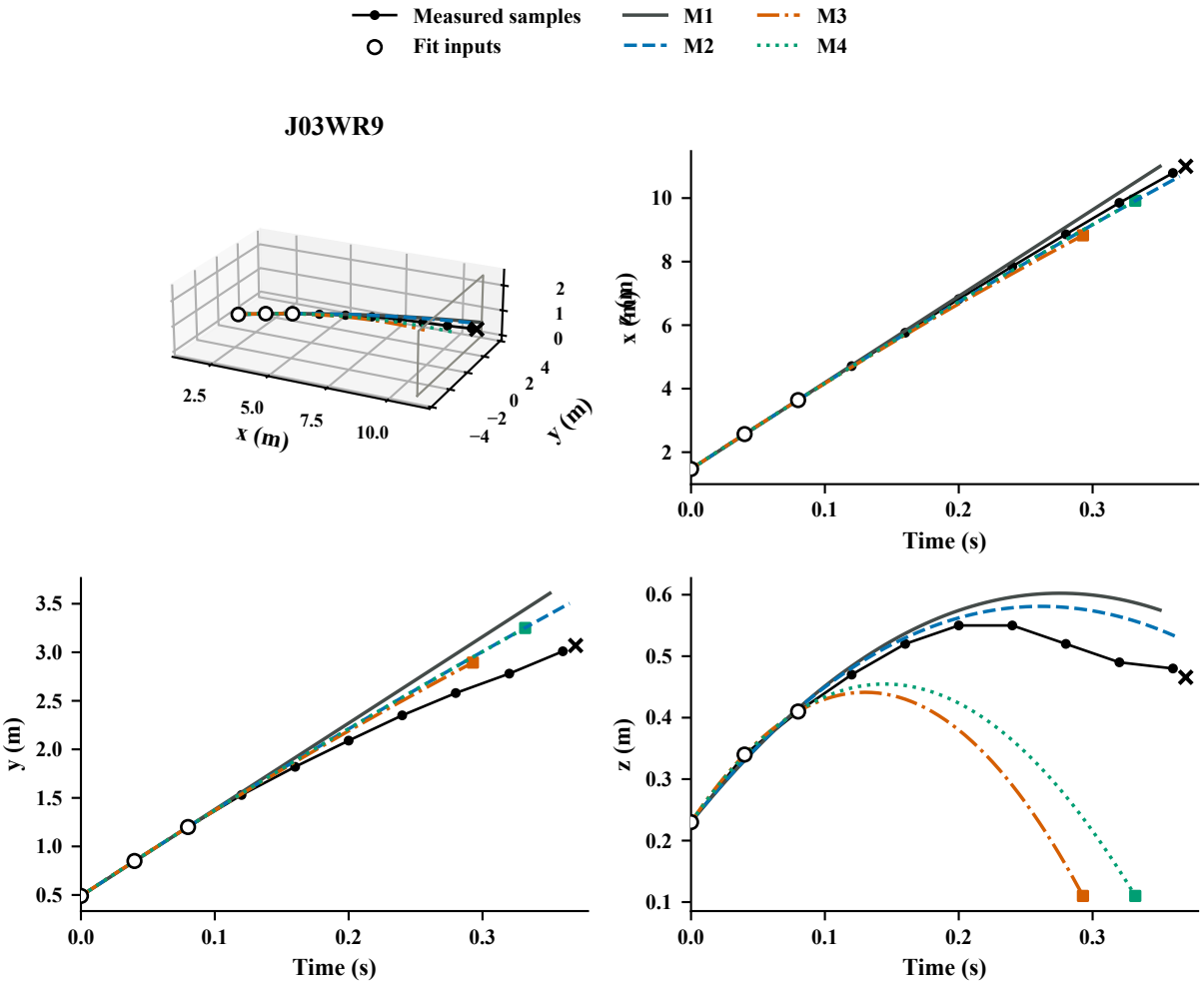


Figure 13: J03WR9: predicted and sampled trajectories from the selected start; fitting inputs are at 0, 0.04 and 0.08 s.

Table 10: J03WR9: inverse initial states and predicted goal-plane crossings. M4 uses a 30 rad/s reference spin magnitude; M1–M3 do not estimate spin.

Model	Status	y_g (m)	z_g (m)	$\boldsymbol{v}_0$ (m/s)	Spin	d_g (m)
M1	plane	3.6083	0.5750	$\begin{pmatrix}27.200\\8.900\\2.703\end{pmatrix}$	—	0.5496
M2	plane	3.6083	0.5147	$\begin{pmatrix}27.747\\9.079\\2.753\end{pmatrix}$	—	0.5408
M3	ground	—	—	$\begin{pmatrix}27.875\\9.125\\3.250\end{pmatrix}$	—	—
M4	ground	—	—	$\begin{pmatrix}27.711\\9.071\\3.170\end{pmatrix}$	$\begin{pmatrix}-9.311\\28.518\\-0.241\end{pmatrix}$	—

The measured path bends laterally, so fitted spin direction affects the continuation towards the goal. M4 reaches the ground at $x = 9.917$ m. Ground contact precedes the goal for M3, M4.

5 Sensitivity Analysis

We retain the individual-parameter, early-position-constrained and Monte Carlo structure for the current M4 equations. The reference observations are $(0, 0, 0.11)$ m at 0 s, $(1.50, 0.45, 0.44)$ m at 0.05 s, and $(2.96, 0.86, 0.74)$ m at 0.10 s. Mass and radius are fixed at 0.43 kg and 0.11 m.

The fitted reference uses $\boldsymbol{v}_0 = (30.3784, 9.4074, 6.8974)$ m/s, $\boldsymbol{\omega}_0 = (19.7434, 1.2060, -22.5553)$ rad/s, $C_D = 0.3623$ and $\alpha = 2.2877$. The crossing is $(2.6159, 1.9526)$ m at 0.398967 s. The 30 rad/s spin magnitude is a reference setting; the maximum early-coordinate residual is 0.000784 m and both coefficients reach the fitted upper bounds.

5.1 Effects of individual parameters

One parameter changes at a time: speed by $\pm 5\%$; spin magnitude, drag coefficient, lift parameter and air density by $\pm 10\%$; velocity and spin-axis angles by $\pm 2^\circ$. Each uses 41 values, totaling 369 flights. The table compares $\pm 1^\circ$ angles and $\pm 5\%$ scalar changes.

Table 11: Goal-plane displacement and arrival-time change under one-factor perturbations.

Parameter	Perturbation	max D_g (cm)	max $\lvert\Delta t_g\rvert$ (ms)
Launch speed	$\pm 5\%$	8.356	20.659
Velocity azimuth	$\pm 1^\circ$	20.070	1.965
Velocity elevation	$\pm 1^\circ$	20.218	1.692
Spin magnitude	$\pm 5\%$	3.873	0.287
Spin-axis azimuth	$\pm 1^\circ$	1.002	0.071
Spin-axis elevation	$\pm 1^\circ$	0.852	0.084
Drag multiplier	$\pm 5\%$	0.620	2.274
Lift multiplier	$\pm 5\%$	3.873	0.287
Air density	$\pm 5\%$	4.142	1.990

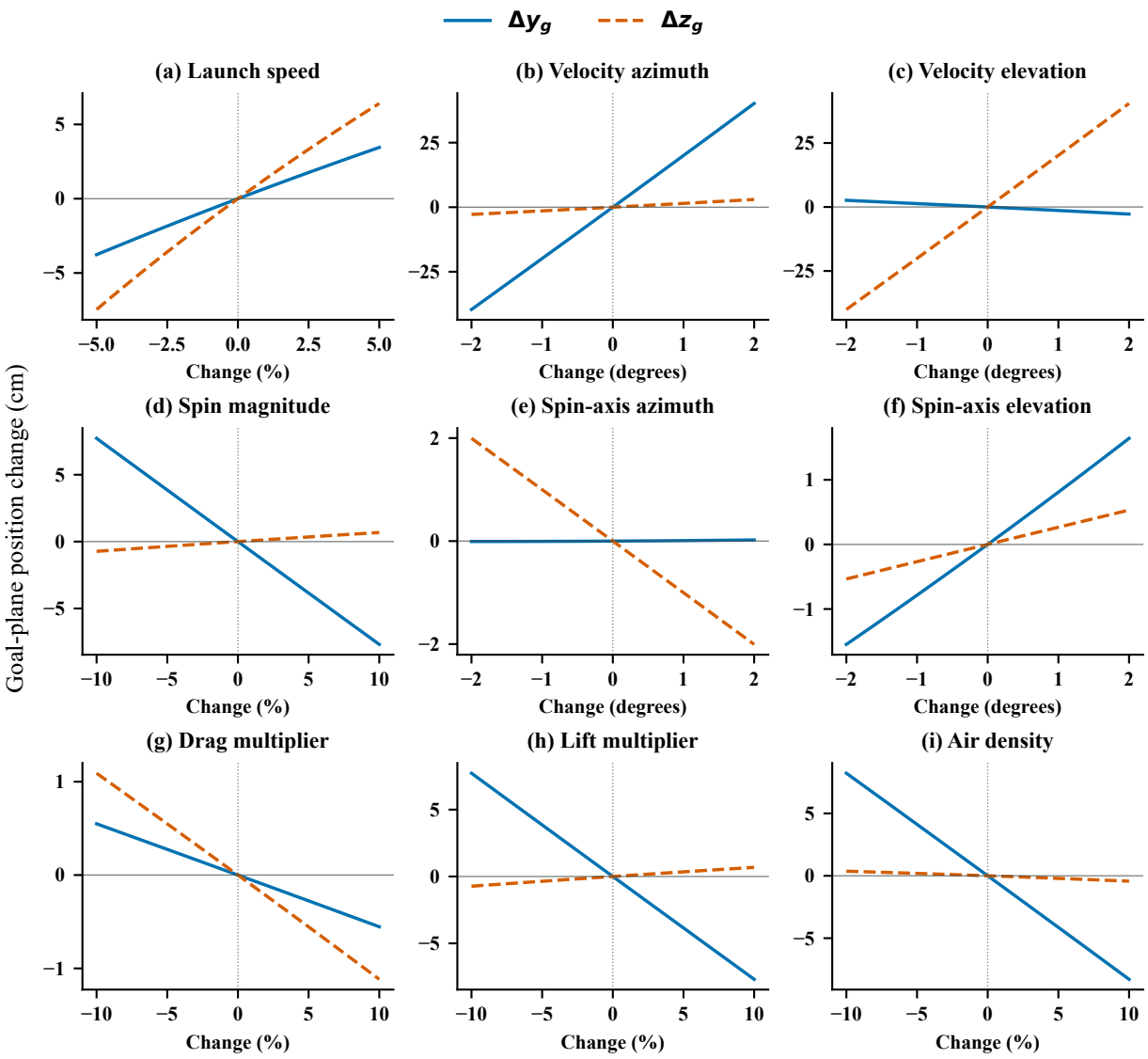


Figure 14: Goal-plane coordinate responses to the nine individual parameters.

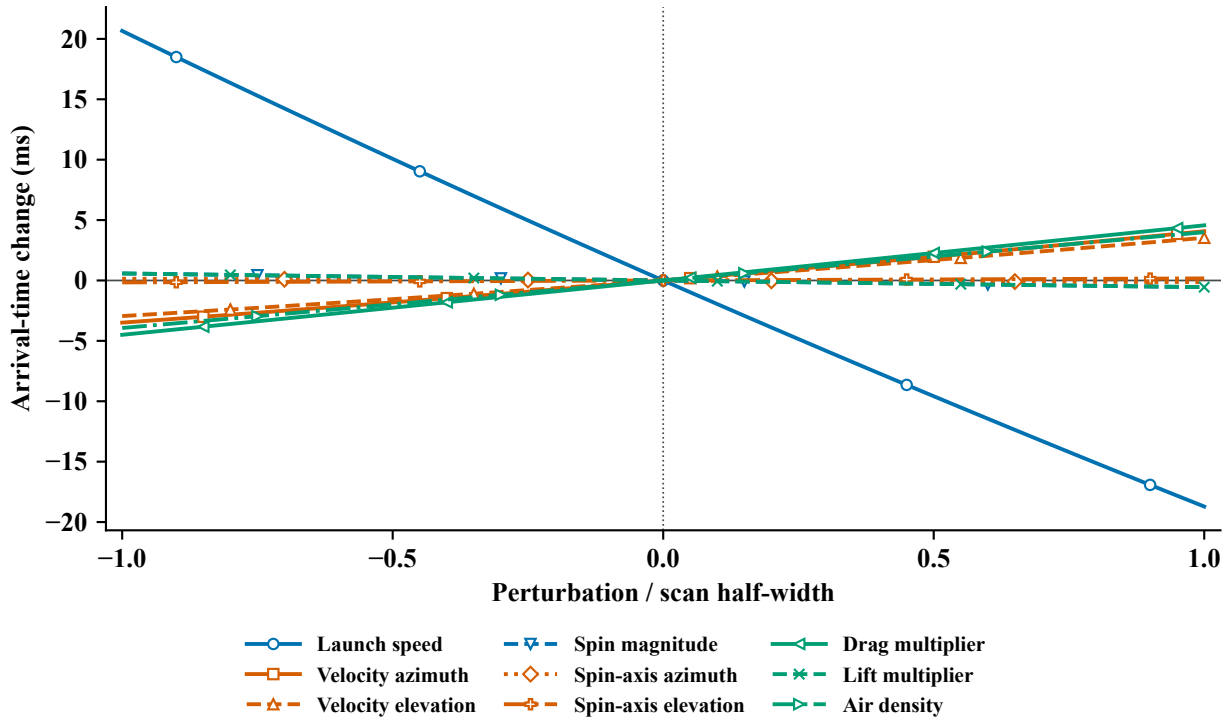


Figure 15: Arrival-time response, with perturbations normalized by each scan half-width.

Spin magnitude and α enter Magnus force as a product, so changes in opposite directions can compensate.

5.2 Crossing variation under early-position constraints

We scan a 17×17 grid of the two coefficient multipliers over $[1.2, 1.8]^2$, refitting velocity and spin components at each node. Total speed is at most 35 m/s, individual spin components at most 60 rad/s in magnitude, and maximum early-coordinate residual at most 0.002 m. The inner range is $[1.35, 1.65]^2$. Reported ranges describe the sampled feasible candidates.

Table 12: Ranges of goal-plane crossings compatible with the early positions.

Range of each multiplier	y_g (m)	z_g (m)	t_g (s)
$[1.35, 1.65]$	2.5538–2.6550	1.9122–2.0307	0.3973–0.4025
$[1.2, 1.8]$	2.5478–2.6601	1.9075–2.0368	0.3973–0.4043

The largest crossing displacement in the wider scan is 0.1052 m.

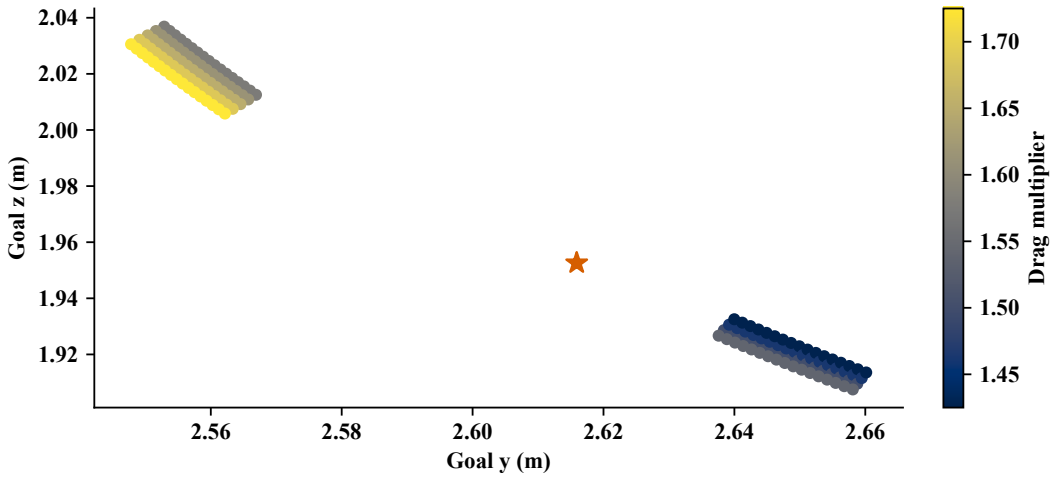


Figure 16: Scanned crossings with early residual at most 2 mm. Colour indicates the drag multiplier.

5.3 Joint perturbations and Monte Carlo simulation

Nine parameters are sampled independently and uniformly over the individual scan ranges, with mass, radius and initial position fixed. Seed 20261108 generates 100000 trajectories; the early positions are allowed to change.

Table 13: Monte Carlo crossing positions and arrival times (100,000 trajectories).

Output	Sample minimum–maximum	Central 95% interval, one coordinate
y_g (m)	1.9633–3.2143	2.1858–3.0505
z_g (m)	1.4381–2.4646	1.5534–2.3535
t_g (s)	0.3685–0.4346	0.3783–0.4218

The crossing mean is $(2.6170, 1.9511)$ m and standard deviations are 0.2455 m and 0.2360 m. Under the ball-centre clearance rule, 95772 trajectories (95.77%) pass fully through the opening and 4228 pass above it.

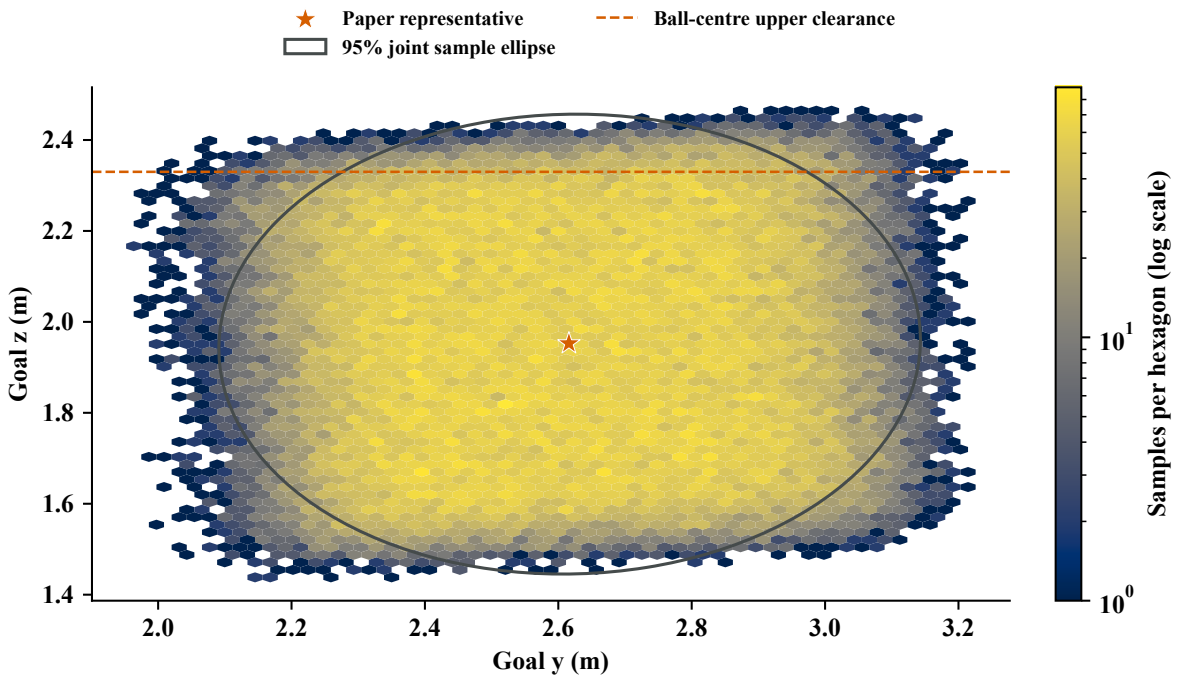


Figure 17: Joint-perturbation crossings; the ellipse covers 95% of simulated samples.

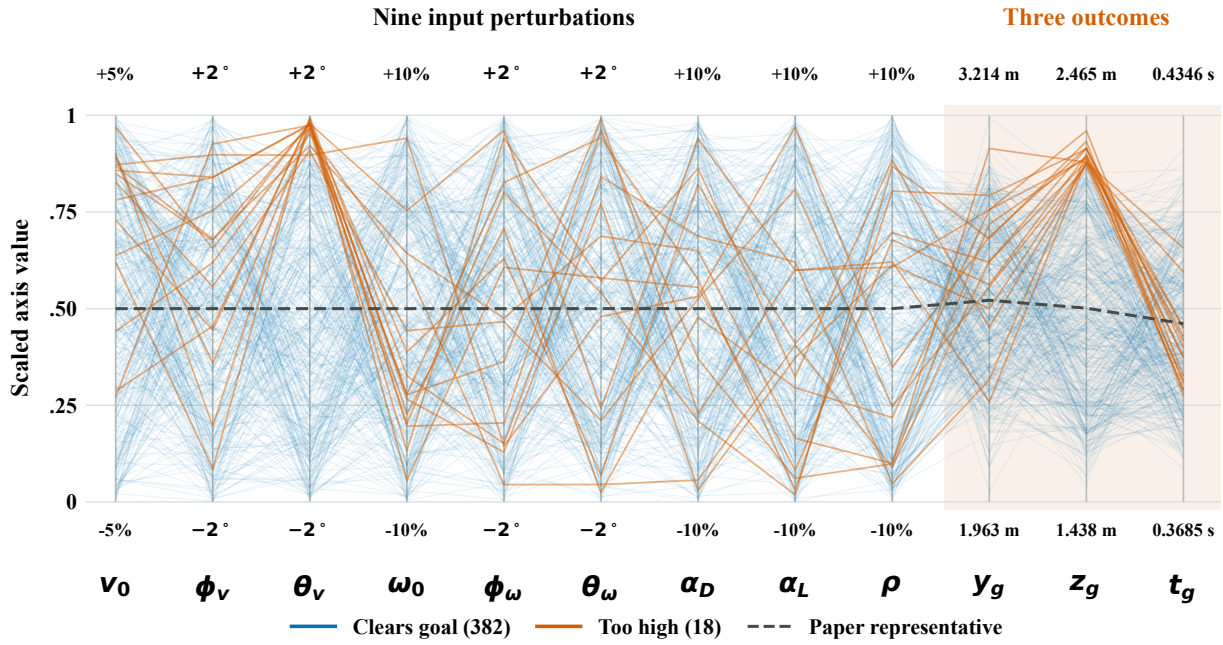


Figure 18: A random display of 400 flights connects the nine inputs to three outputs.

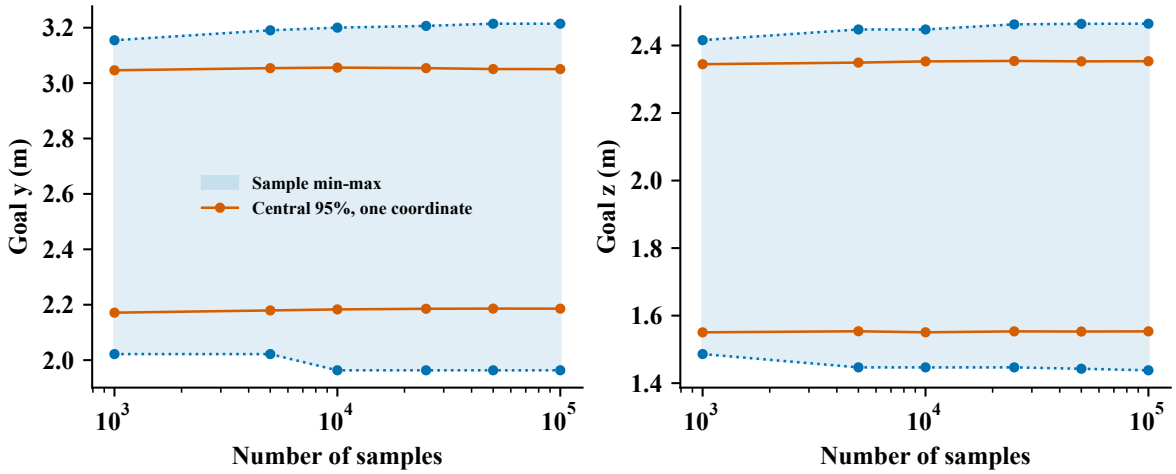


Figure 19: Effect of sample count on the simulated crossing distribution.

Fourth-order Runge–Kutta integration is compared with adaptive integration and a halved step. These ranges and fractions correspond to the stated uniform sampling assumptions.

6 Discussion of Other Factors

The main model isolates gravity, drag and Magnus lift. We now examine the physical effects omitted from its geometry, spin evolution and ambient airflow.

6.1 Asymmetric Surface Geometry

Paudel and Yan [9] performed full-flight flow simulations using realistic football geometry and a smooth-sphere comparison. For launch speeds of 10, 15 and 20 m/s and spin rates of 31.5, 63 and 94.5 rad/s, their results show that seams and surface structure alter separation, wake dynamics and Magnus force. The smooth sphere produces greater lateral bending in their nine comparisons, and vertical flight-time differences become more visible at higher speed and rotation.



Figure 20: Illustrative comparison of a paneled football and a smooth sphere, inspired by Paudel and Yan [9]. Their compared bodies had diameter 0.22 m and mass 0.436 kg. The panel geometry shown here is schematic.

Alam et al. [16] measured drag coefficients over a range of Reynolds numbers for six footballs and a smooth sphere. Figure 21 shows the panels and surface textures of these footballs, while Figure 22 compares their drag coefficients. The differences illustrate the influence of ball surface structure on aerodynamic drag.

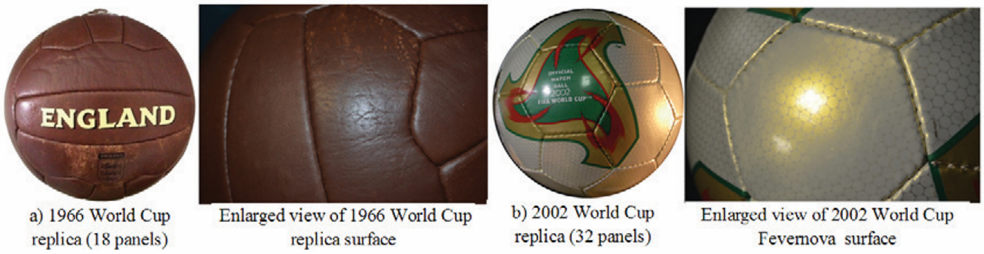


Fig. 1. Vintage and Adidas Fevernova balls



Fig. 2. Adidas Teamgeist 2006 and 2009 balls



Fig. 3. Adidas 2010 Jabulani and Nike 2010 T-90 balls

Figure 21: Six footballs and details of their surfaces, reproduced from Figures 1–3 of Alam et al. [16].

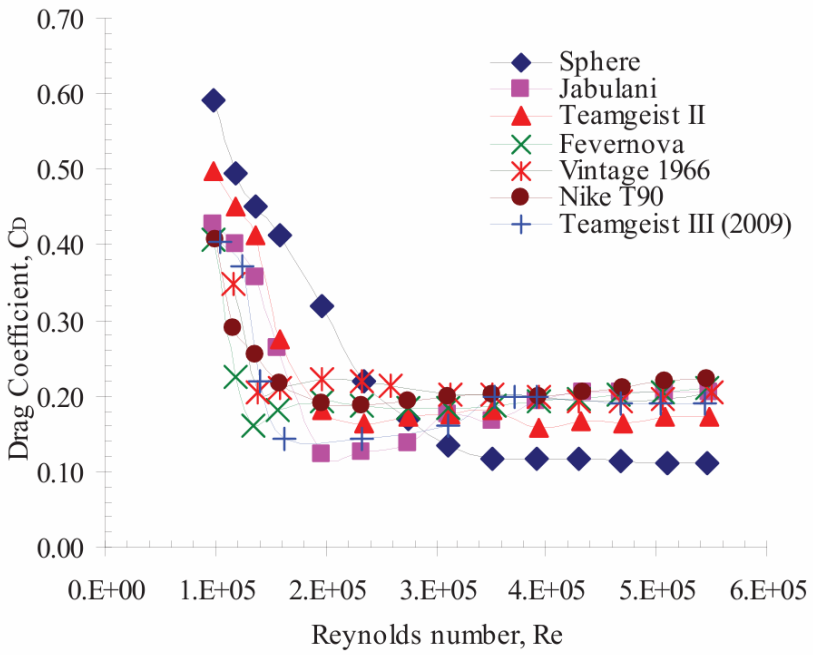


Figure 22: Drag coefficient against Reynolds number for different footballs and a smooth sphere, reproduced from Figure 5 of Alam et al. [16].

The spherical geometry in the present report describes the centre-of-mass motion and projected area. Its force coefficients come from football measurements, so the average surface influence enters through C_D and C_L. Explicit panel orientation and asymmetric wake fluctuations provide a further level of description when lateral-force details are important. These published results support retaining football-specific coefficients in a spherical trajectory model.

6.2 Rotation and Aerodynamic Damping

Rotation changes Magnus force, so a declining spin can change the continuation of a curved shot. According to Grad et al. [13], Supplementary Section 8.4, and the MuJoCo ellipsoid model [14], the spherical inertial and viscous damping torques are

$$\boldsymbol{Q} = -k_2\omega\boldsymbol{\omega} - k_1\boldsymbol{\omega}, \quad k_2 = \frac{8\pi}{15}\rho C_{\text{angular}} R^5, \quad k_1 = 8\pi\mu R^3. \tag{12}$$

The quadratic term describes high-Reynolds-number angular drag; the linear term is viscous resistance. We use MuJoCo's documented reference value $C_{\text{angular}} = 1.5$ [14]. Defining $q = \rho v^2/2$ and $C_{Q,v} = \|\boldsymbol{Q}\|/(qAR)$ gives

$$C_{Q,v} = \frac{16}{15} C_{\text{angular}} Sp^2 + \frac{32Sp}{Re}, \qquad C_{Q,\omega} = \frac{C_{Q,v}}{Sp^2} = \frac{16}{15} C_{\text{angular}} + \frac{32}{Re\, Sp}. \tag{13}$$

$C_{Q,\omega}$ uses rotating surface speed $R\omega$ in its reference pressure. The factor $1/Sp^2$ converts between the two normalizations. The viscous-to-inertial torque ratio is

$$\frac{Q_{\text{viscous}}}{Q_{\text{inertial}}} = \frac{30}{C_{\text{angular}} Re\, Sp} = \frac{15\nu}{C_{\text{angular}} R^2 \omega}. \tag{14}$$

It is 3.54×10^{-4} at the prescribed curved-shot launch, so the quadratic term dominates. Following the thin-shell inertia used by Maponi and Piombin [8], $I_b = 2mR^2/3$. Adding the high-Reynolds-number torque gives

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ I_b\dot{\boldsymbol{\omega}} = -k_2\omega\boldsymbol{\omega}. \end{cases} \tag{15}$$

The torque opposes spin and removes rotational energy. With this isolated quadratic law, $\omega(t) = \omega_0/[1 + (k_2/I_b)\omega_0 t]$, providing an integration check.

Use the curved shot of Section 4.1, with initial velocity $(23.6354, 0, 4.1676)$ m/s, spin $(0, 0, 35)$ rad/s and unit force corrections. Holding this state fixed, we compare constant spin with Eq. (15).

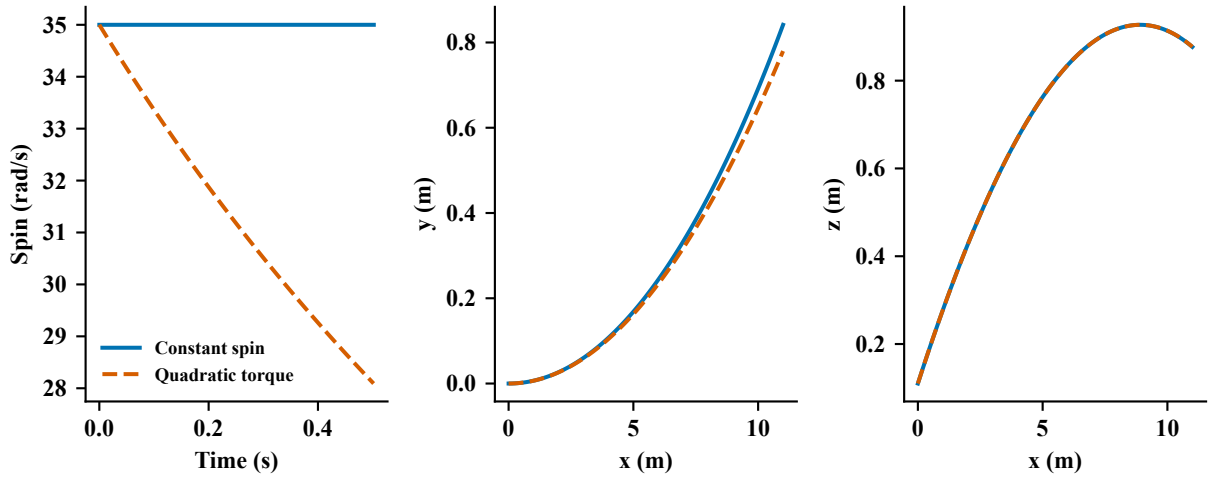


Figure 23: Same-initial-state comparison of constant spin and high-Reynolds-number rotational damping for the prescribed curved shot. The documented angular-drag reference is 1.5, and the ball uses thin-shell inertia.

The damped crossing is $(0.7799, 0.8772)$ m, compared with $(0.8415, 0.8769)$ m at constant spin. The displacement is 0.0616 m (0.560% of 11 m). Spin falls by 19.740%, and crossing time changes by -0.066%. Adding viscous torque shifts the damped crossing by 2.1e-05 m. The short-flight endpoint is less sensitive than spin magnitude to this reference torque.

6.3 Wind Speed and Force Estimates

Wind changes the airflow through $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$ [8]. At fixed coefficients, a wind speed w along the flight direction changes the dynamic-pressure scale by

$$\frac{q_{\text{head}}}{q_0} = (1 + w/v)^2, \qquad \frac{q_{\text{tail}}}{q_0} = (1 - w/v)^2, \qquad \frac{q_{\text{cross}}}{q_0} = 1 + (w/v)^2. \tag{16}$$

Crosswind also rotates the airflow by $\arctan(w/v)$, changing the force direction. Table 14 evaluates these rough estimates at the curved-shot launch speed. The listed wind speeds are illustrative conditions for comparison.

Table 14: Illustrative wind at the 24 m/s curved-shot launch: dynamic-pressure changes at fixed coefficients.

w (m/s)	Headwind (%)	Tailwind (%)	Crosswind (%)	Angle (°)
1	+8.51	-8.16	+0.17	2.39
3	+26.56	-23.44	+1.56	7.13
5	+46.01	-37.33	+4.34	11.77

A 5% limit on the headwind dynamic-pressure change corresponds to $w/v \leq \sqrt{1.05} - 1$, or $w \leq 0.59$ m/s at this launch. The still-air approximation is most suitable for indoor or weak-wind flight; stronger wind should enter through the relative-velocity calculation.

7 Conclusions

The model separates the mechanisms of gravity, drag and Magnus lift. Gravity produces downward curvature, drag reduces speed, and spin redirects the path. We use constant drag and linear-spin lift.

Five prescribed kick situations provide six initial states and 24 forward flights. The matched top- and backspin pair demonstrates the reversal of vertical Magnus force, while sidespin produces horizontal curvature. In the curved M4 example the goal crossing is $(0.8415, 0.8769)$ m at 0.5022 s. The same initial state is used to quantify angular damping, giving a crossing displacement of 0.0616 m under the cited reference setting.

Three actual penalties supply distinct early sampling sequences and held-out trajectories. Coordinate conversion places all three in the competition geometry. The 12 inverse outcomes list fitted initial states, goal crossings, crossing deviations and ground termination separately, connecting the recorded motion to each force assumption. For the competition-position example in Section 5, a one-degree launch-direction change moves the crossing by the displacement shown in the one-factor table. With the early positions fixed, the explored compatible states shift the crossing by up to 0.1052 m; joint perturbations are quantified by 100,000 Monte Carlo trajectories with fixed ball mass and radius.

8 Model Evaluation

8.1 Physical Interpretation and Reproduction

The four-model hierarchy makes the role of each force visible under shared initial conditions. Forward and inverse calculations use the same state derivative and coefficient definitions. Original source coordinates, selected frames, transformed coordinates, numerical settings and candidate solutions are saved with the results.

8.2 Prediction and Further Development

The real-penalty comparison evaluates how early motion carries into the later path. Bound-active solutions and residuals identify the influence of the chosen coefficient scale and initial-state search. Ball-specific coefficients, direct spin measurements and higher-rate synchronized tracking would improve these connections. Rotational damping and wind can be included when their effect is relevant to the flight considered.

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Reproduction Code

The project contains the original penalty CSV, coordinate conversion, shared inversion, prediction results and figure scripts. The excerpts show the common state equation and force evaluation. Paper and program use the same coefficients and bounds; computation, plotting and bilingual table generation are separate commands.

```
--
    return -factor*w


def state_derivative(t, state, config):
    """The single nine-state RHS used for prediction and inverse fitting."""
```

```
--
    return record


def forces(velocity, spin, config):
    """Return force components on ball, in newtons."""
    v = np.asarray(velocity, dtype=float) - np.asarray(config.wind, dtype=float)
    omega = np.asarray(spin, dtype=float)
    speed = np.linalg.norm(v)
    weight = (config.mass * np.asarray(config.constant_acceleration, dtype=float)
              if config.constant_acceleration is not None else
              np.array([0., 0., -config.mass * config.gravity if config.use_gravity else
                  0.]))
    drag = np.zeros(3)
    magnus = np.zeros(3)
    if speed > 1e-12:
        cd, cl, _, _, _ = coefficients(velocity, spin, config)
        q_area = 0.5 * config.air_density * np.pi * config.radius**2 * speed**2
        if config.use_drag:
            drag = -q_area * config.drag_scale * cd * v / speed
        cross = np.cross(omega, v)
        norm = np.linalg.norm(cross)
        if config.use_magnus and norm > 1e-12:
```